Chemistry (joint degree)

Vrije Universiteit Amsterdam - Faculty of Science - Chemistry (joint degree) - 2025-2026

Chemistry is a joint degree master's programme of University of Amsterdam and VU Amsterdam. For further information about the programme go to the UvA Course Catalogue:

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-opleiding/opleiding/8801

Info

Level	Master
Language	English
Duration	2 years
Type	Full and Part Time
Credits	120 EC
Faculty	Faculty of Science

More information about this programme

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Chemistry joint degree Constrained choice couses taught at VU

Courses

Course Name	Period	Credits	Code
Bioanalytical Sciences	P1	6EC	X_432765
BioSolar Cells	P1	6EC	X_428531
Physical Organic Chemistry	P1	6EC	X_435663
Chemtech-to-Market Value Creation Lab	P2	6EC	XMU_0067
Management of Sustainable Innovation	P2	6EC	E_DBI_MANSI
Plastic and Chemical Pollution	P2	6EC	AM_1259
Understanding Quantum Chemistry	P2	6EC	X_422557
Analytical NMR Spectroscopy	P4	6EC	XMUJ_0010
Molecular Modelling	P4	6EC	XMU_0059
Biocatalysis for a Sustainable Future	P5	6EC	XMU_0041
Photosynthesis and Energy	P5	6EC	X_422553
Protein Analysis	P5	6EC	X_435045
Synthetic Strategies	P5	6EC	XMU_0016
Advanced Computational Chemistry	P6	6EC	XMU_0014
Advanced Spectroscopy	P6	6EC	X_432767
High-Throughput Screening	P6	6EC	X_435047

Chemistry JD Compulsory Courses taught at VU

Courses

Course Name	Period	Credits	Code
Separation Sciences	P1	6EC	X_435609
Current Sustainable Energy Technologies	P3	6EC	X_422582
Concepts in Chemical Biology	P4	6EC	XM_0080
Spectroscopic Analysis	P5	6EC	XMU_0039

Chemistry JD Optional Courses taught at VU

Courses

Course Name	Period	Credits	Code
ADME	P1	6EC	XM_0136
Biotechnology for Molecular Discovery	P1	6EC	XMU_0043
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Advanced Molecular Orbital Theory	P4	6EC	XMU_0028
Molecular Modelling	P4	6EC	XMU_0059
Relativistic Quantum Chemistry	P4	6EC	XMU_0029
Scientific Computing and Programming	P4	6EC	X_435076
Science and Society in Historical Perspective	P4+5	6EC	X_400424
Advanced Biophysics	P5	6EC	X_422583
Second Quantization Techniques in Quantum Chemistry	P5	6EC	XMU_0030
Synthetic Strategies	P5	6EC	XMU_0016

Chemistry JD Suggested Academic Skills Courses taught at VU

Courses

Course Name	Period	Credits	Code
Teaching Assistant	Ac. Year (September)	3EC	XM_432741
Teaching Assistant	Ac. Year (September)	6EC	XM_432742
Science and Communication	P1	6EC	AM_470587
Communication, Organization and Management	P2	6EC	AM_470572
Science in Dialogue	P3	6EC	AM_1002
Tutoring Students	P3	3EC	X_432625
Scientific Writing in English	P3, P6	3EC	X_400592
Entrepreneurship in Science and Technology	P4	6EC	XMU_0066

Courses

ADME

Course Code	XM_0136
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Moco
Examiner	dr. S. Moco
Teaching Staff	dr. S. Moco, dr. J. Kool
Teaching method(s)	Study Group, Other, Lecture

Course Objective

“What happens to a drug inside the human body?”

This course will dive into the biochemical processes underlying the absorption (A), distribution (D), metabolism (M) and excretion (E) of drugs (or xenobiotics, in general): the ADME processes, an essential discipline in pharmaceutical sciences.

Course Content

Of the thousands of novel compounds that are developed in drug discovery programs, only a fraction has the appropriate pharmacokinetic properties to become a drug product. Pharmacokinetics is determined by the complex processes involved in absorption (A), distribution (D), metabolism (M) and excretion (E) of the drug in the human body, the so-called ADME processes. ADME processes directly impact drug effectiveness (pharmacodynamics), safety and dosage. ADME processes do not only apply to pharmaceutical drugs, but environmental contaminants, nutraceuticals or phytochemicals may follow analogous processes. In this course, the student will learn: the fundamentals of ADME, state-of-the-art analytical chemistry approaches to study ADME, and inter-individual variability impacting ADME.

In this course, the students will be familiarised with:

Drug **A**bsorption

- routes of administration, with emphasis on the gastro-intestinal tract (GIT);
- drug transporters involved in absorption;
- how much drug reaches systemic circulation

Drug **D**istribution

- how does the drug spread through the body’s tissues and fluids;
- blood brain barrier;
- drug protein binding and volume of distribution

Drug **M**etabolism

- how does the body chemically change the drug;
- xenobiotic metabolism mostly in the liver and gut, mediated by major enzymes classes and isoforms: CYPs, UGTs, SULTs, etc, and integrated metabolic pathway maps;
- (re)active metabolites and drug-drug interactions

Drug **E**xcretion

- how does the body eliminate the drug (often via urine or bile);
- drug transporters involved in excretion;
- drug clearance and entero-hepatic recirculation

Methods and approaches to study ADME processes

- biochemical, cellular, pre-clinical and clinical studies;
- analytical chemistry tools to monitor drugs and metabolites thereof in biological samples: LC-MS and NMR metabolomics;
- strategies to perform drug metabolite identification (metID) from LC-MS/MS;
- bioanalytical strategies for bio-affinity profiling of metabolic mixtures;
- use of (stable) isotopically labelled drugs to study ADME

Factors impacting ADME processes

- physico-chemical properties of drugs;
- inter-individual variability in: environmental factors, genetic variants (metabolic enzymes, transporters), sub-populations;
- personalised medicine

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Written exam and assignments

Literature

Consulting books:

1. ‘ADME Processes in Pharmaceutical Sciences’. Eds. Alan Talevi, Pablo A. M. Quiroga. Springer 2018, ISBN 978- 3-319-99593-9 2.
2. 'Rang & Dale's Pharmacology’. Eds. J.M. Ritter and R.J. Flower. Elsevier 2023, ISBN: 978-0-3238-7395-6.
3. ‘Drug-like properties: concepts, structure design and methods. From ADME to Toxicity optimization.’ Eds. E.H. Kerns and L. Di, Academic Press, 2008, ISBN: 978-0-1236-9520-8.

Additional course material will be provided in Canvas.

Additional Information Target Audience

All MSc DDS specialisations, MSc Biomolecular Sciences and MSc Chemistry (tracks Chemistry of Life and Analytical Sciences)

Recommended background knowledge

Background knowledge in chemistry is recommended.

Advanced Biophysics

Course Code	X_422583
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Dulin
Examiner	dr. D. Dulin
Teaching Staff	dr. D. Dulin, dr. ir. I. Heller

Teaching method(s)	Lecture

Course Objective

Learning objectives

- Understand the physics underlying the biological problems and biophysical research methods featured in this course
- Be able to explain how these methods can be exploited to address specific biological problems and identify strengths and weaknesses of methods in biological context
- Critically reflect on your understanding of methods application in assigned literature and formulate relevant questions
- Analyze scientific literature that uses the discussed methodologies to addresses biological problems and reflect on suitability of used methods
- Communicate and present your literature analysis and critically reflect on analyses by other students

Course Content

Life is, by its very definition, a dynamic quantity. In this course an overview is given of current biophysics, biophotonics, and biomedical research fields that investigate the dynamic processes that take place in biomolecules, membranes, cells and tissues and are indispensable for biological functionality. This course aims to highlight the 'what', 'why' and 'how' of these research fields: what are the established and new biophysical insights, why are these insights important and interesting, and how are these results being obtained.

The following biophysical/biomedical topics and techniques are featured in this course:

- DNA processing and dynamics
  o Topics: DNA structure and stability, DNA/RNA polymerase, DNA architectural proteins, DNA repair.
  o Techniques: optical tweezers, AFM, tethered particle motion, magnetic tweezers, superresolution imaging.
- Protein dynamics
  o Topics: Photoactive proteins, light-driven enzymes, Motor proteins, optogenetics.
  o Techniques: infrared spectroscopy, time-resolved spectroscopy, laser techniques
- Membrane dynamics
  o Topics: Photosynthesis, membrane protein diffusion, Neuroimaging.
  o Techniques: AFM, single molecule, nonlinear imaging, polarized spectroscopy.
- Cell/tissue dynamics
  o Topics: Optical and/or mechanical characterization of tissue or cells.
  o Techniques: (Fluorescence) microscopy, optical coherence microscopy, mechanical analysis.

Additional Information Teaching Methods

Lectures, guest lectures, literature essay and assignments, oral literature presentation

Method of Assessment

- Literature assignments
- Essay (literature or research proposal) (70% final grade)
- Essays from two other students peer review (10% final grade)
- Oral literature presentation (20% final grade)

Entry Requirements

BSc. Physics, BSc. Medical Natural Sciences, BSc Chemistry or comparable.

For MNS students to participate in this course, it is advised to have successfully completed the course Microscopie en spectroscopie voor MNW (X_420586) and Mechanics and Thermodynamics in the Cell (X_422589).

Literature

Notes, handouts and papers.

Additional Information Target Audience

mMNS-PoL, mPhys-LSBP, mPhys-PLH

Advanced Computational Chemistry

Course Code	XMU_0014
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. F.M. Bickelhaupt
Examiner	
Teaching Staff	dr. T.A. Hamlin, prof. dr. C. Fonseca Guerra, prof. dr. F.M. Bickelhaupt
Teaching method(s)	Computer lab, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130546

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Advanced Molecular Orbital Theory

Course Code	XMU_0028
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. F.M. Bickelhaupt
Examiner	prof. dr. F.M. Bickelhaupt
Teaching Staff	dr. P. Vermeeren, prof. dr. C. Fonseca Guerra, prof. dr. F.M. Bickelhaupt
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130551

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Advanced Spectroscopy

Course Code	X_432767
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A. Bhulai
Examiner	dr. A. Bhulai
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

Learning objectives:

- To deepen understanding obtained during earlier courses in the field of spectroscopy
- To acquire understanding of the principles of advanced spectroscopy modes
- To obtain insight into the practical and instrumental aspects of modern molecular spectroscopy; to further develop practical skills
- To practice extracting information from the scientific literature
- To further develop presentation skills
- To further develop scientific writing skills

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130533

Additional Information Teaching Methods

- Lectures/ tutorials
- practical training at LaserLaB VU
- Presentations, reporting
- Collaboration in small groups
- Self-study

Method of Assessment

The final grade will be determined based on Experiment report, Experiment presentation, Literature presentation, Written exam.

Entry Requirements

Background knowledge of molecular spectroscopic techniques, in particular fluorescence and Raman, is expected (e.g. MSc Chemistry course Spectroscopic Analysis or 3MNW course Microscopie en Spectroscopie).

Literature

will be made available via Canvas

Additional Information Target Audience

Aimed at students in the MSc Chemistry or MSc Biomedical Technology & Physics programs, but also open to interested students from other programs with the necessary background in molecular spectroscopy

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Recommended background knowledge

Background knowledge of molecular spectroscopic techniques, in particular fluorescence and Raman, is expected (e.g. MSc Chemistry course Spectroscopic Analysis or 3MNW course Microscopie en Spectroscopie).

Analytical NMR Spectroscopy

Course Code	XMUJ_0010
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Moco
Examiner	dr. S. Moco
Teaching Staff	dr. T.A. Hamlin, dr. S. Moco, A.C. Janssen
Teaching method(s)	Excursion, Seminar, Computer lab, Symposium, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130544

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Bioanalytical Sciences

Course Code	X_432765
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I. Kohler
Examiner	dr. I. Kohler
Teaching Staff	dr. I. Kohler
Teaching method(s)	Lecture

Course Objective

Get familiar with the bioanalytical techniques and strategies used in clinical and toxicological fields.

Be capable of developing a bioanalytical strategy based on a clinical/toxicological question, from the sampling to the data analysis and interpretation.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130555

Additional Information Teaching Methods

1. Lectures, including case studies: during the lectures, the students will learn about the analytical strategies implemented in the field of bioanalysis, including sample collection, sample storage, sample preparation, separation of analytes, detection, and data analysis/interpretation. The lectures focus on relevant bioanalytical applications in the field of clinical and forensic toxicology, therapeutic drug monitoring, and biomarker discovery. By focusing on the application and on real-life case studies, students will better understand the requirements needed for the development of a suitable method, which will help them understand which choices should be made for each step of the bioanalytical workflow, and why. Two Lectures per week; each lecture is 2x45 min.
2. Group assignment: during the group assignment, performed in small groups (max 6 students/group), students will propose an analytical strategy suited for the qualitative or quantitative analysis of endogenous or exogenous compounds in a complex biological matrix. Students will have to think about the biological/clinical question (why?), the compounds to be measured (what?), and the analytical workflow they need to develop to measure those compounds (how?). This group assignment will help students (i) improving their understanding of the concepts presented during the lecture, (ii) stimulate their scientific curiosity, (iii) extend their literature search skills, and (iv) train their presentation and writing skills. One block of self-study scheduled every week (2x45 min)
3. Online Quizzes: students will practice their newly acquired knowledge every week via online quizzes, which contain exam-like questions. Quizzes can be taken during the self-study time or at home at a later stage.

Method of Assessment

Group assignment: poster, presentation and video (group grade)

Lectures and Quizzes: written exam (individual grade)

Entry Requirements

Basic knowledge of biochemistry, chromatography, electrophoresis and mass spectrometry

Literature

Powerpoint slides, additional articles provided during the course, and tutorial videos to support the students for the group assignment

Additional Information Target Audience

mCH-AS, mCH-CoL, mMNS, mFS. The course is not available for mDDS-Bioanalytics

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Recommended background knowledge

Basic knowledge of biochemistry, chromatography, electrophoresis and mass spectrometry.

Biocatalysis for a Sustainable Future

Course Code	XMU_0041
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	dr. I. Drienovska
Examiner	
Teaching Staff	dr. I. Drienovska
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130563

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

BioSolar Cells

Course Code	X_428531
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.N. Frese
Examiner	dr. R.N. Frese
Teaching Staff	dr. R.N. Frese
Teaching method(s)	Lecture

Course Objective

The course provides a broad overview of BioSolar Cells scientific research. It serves to acquire detailed knowledge of parts that comprise biological, chemical and physical materials approaches to utilize sunlight for fuel and or capture carbon dioxide based on biological or artificial photosynthesis.

Focus is on technologies that are being researched right now and are not yet implemented at full scale. Part of the course is dedicated to technology transition theory, focusing on the hurdles and bottlenecks for lab-based researched to be implemented. Important skills that are developed are the assessment of technologies that are not yet implemented at scale for current decision makers in renewable energy.

In the course, students will be reading scientific papers from the exact sciences and other academics, information leaflets for different target audiences on renewable energy efforts. Subsequently, students will be following lectures on focused topics and learn to distill relevant information from them. In work, students aim to present an overview of specific technology using scientific and other references with an assessment of the relevance of references used.

Combined this course will allow students to investigate a topic of their own interest and to acquire the necessary skills of assessing and presenting scientific information. The course provides the basic skills for a successful master study program.

The course consists of

- a group assignment on a relevant issue related to Biosolar Cells, presentation
- a literature study on one of the themes from BioSolar Cells, using a template geared to fast and critical information transfer
- a set of questions and annotations in a diverse set of reading materials

Course Content

The course consists of three parts, and starts with interactive lectures by experts on each of the three themes from BioSolar Cells. Students will read one or two papers before each lecture and formulate questions that have been researched by themselves first. In the second stage of this course, the students first work out a group assignment (groups of 3-4 students) leading to a team presentation on a specific topic, and then conduct an individual assessment on a BioSolar-Cells related topic of their own choice and present their work in the form of a scientific review paper where the specific topic is compared with other solutions, the paper concludes with a brief description of a research proposal.

Additional Information Teaching Methods

- Lecture
- Self-study

Method of Assessment

Grade is a weighted sum of the presentation and the report.

For the quality of the questions submitted prior to the lectures the student will receive the judgement “insufficient”, “sufficient” or “excellent” by the lecturer and a +/- bonus will be added/subtracted from the grade depending on the final sum.

Also for the quality of the referee reports the student will receive the judgement “insufficient”, “sufficient” or “excellent” and +/- bonus point. Maximum grade is always 10.

A retake is only allowed once and only when all materials have been submitted with requested content and within the prescribed deadline.

Entry Requirements

Physics or chemistry at bachelor level or a mixture of these at bachelor level supplemented with other topics. Highschool biology is recommended. For those without prior knowledge on photosynthesis an introductory lecture and reading material will be provided.

Literature

During the course a diverse set of scientific papers, leaflets, academic papers, brochures, and other materials is provided for reading and annotating (compulsory)

Additional Information Target Audience

master students physics, chemistry, SBI, lifesciences

Additional Information

This course can be challenging for those that do not read much.

Explanation Canvas

Information, materials, rubrics and templates are on canvas, provided on a weekly basis. Students are expected to read all materials without specific instructions from a teacher.

Recommended background knowledge

for those without prior knowledge on photosynthesis an introductory lecture will be given

Biotechnology for Molecular Discovery

Course Code	XMU_0043
Credits	6

Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.A.K. Jongkees
Examiner	dr. S.A.K. Jongkees
Teaching Staff	dr. S.A.K. Jongkees
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130567

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Chemtech-to-Market Value Creation Lab

Course Code	XMU_0067
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Hasanefendic
Examiner	dr. S. Hasanefendic
Teaching Staff	dr. S. Hasanefendic
Teaching method(s)	Study Group, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130575

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Communication, Organization and Management

Course Code	AM_470572
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C.A.C.M. Pittens MSc
Examiner	dr. C.A.C.M. Pittens MSc
Teaching Staff	dr. S.S. Rai, dr. R.M.H. Peters, dr. C.A.C.M. Pittens MSc
Teaching method(s)	Study Group, Lecture

Course Objective

After completion of this course, you will be able to:

- identify and discuss knowledge of, and theories on organisational structures, organisational behavior, organisational change, and change management;
- describe and conceptualise organisations as embedded in a wider system, and interpret these from a transdisciplinary perspective;
- obtain a deeper understanding of communication from the perspective of sharing and influencing results, and to learn how to deal with conflicts;
- identify and discuss theories on leadership and interpersonal behaviour;
- make an informed choice between statistical techniques commonly used in health and life sciences, applying these to conduct analyses while using statistical software (e.g., SPSS) and interpreting the results of these techniques (e.g., t-tests, chi-square tests) to inform decision making;
- practise analytical and advisory skills.

Course Content

Organisations need to respond and adapt to internal and external challenges. It is increasingly recognized that addressing these challenges is not an ‘isolated process’. Organisations (need to) collaborate with multiple stakeholders in these processes of (continuous) change. In-depth understanding of the structure of an organisation and the wider system it operates in is therefore important. Different forms of knowledge can be used to gain this understanding. It requires insights gained from the analyses of qualitative and quantitative data collected within organisations and their wider context. This course offers insights into these complexities, as well as practical (quantitative) methods to address these.

In interactive lectures, several (guest)speakers will discuss macro-aspects of organisational behavior, including designing organisations and managing them, thereby touching on the managerial skills and the strategic thinking required. In addition, concepts related to motivation, managing interpersonal behaviour, leadership, communication and developing and changing organisations will be explained. Furthermore, lecturers will discuss concepts of systems thinking, and the value of adopting multiple perspectives in analysing organisations and processes of organisational change. The (guest) speakers will illuminate concepts derived from the literature by relating them to their own experiences with informing and managing organisational change in practice. Furthermore, the role of quantitative data in managerial decision making will be explored (the role of qualitative data is covered in other courses). To that end, you will explore and discuss different statistical techniques (e.g., t-tests, chi-square tests) commonly used in health and life sciences. You will conduct analyses with these techniques in statistical software (e.g., SPSS), interpret the results and translate these to decision making processes, working towards an advisory report for a (fictive or real-world) organisation’s management.

In an assignment that runs through the course, practical cases of healthcare companies will be analysed and discussed, resulting in advisory reports for management. With the other students you discuss your experiences and a coach helps you relate the experiences to theory.

Additional Information Teaching Methods

Lectures: approximately 22 hours
Response lectures: 4 hours
Training workshops: 16 hours (attendance is mandatory)

Self-study and writing project assignment: remaining hours.

Method of Assessment

Written exam (60%) and assignment (40%). Grades of both components must be sufficient.

Literature

To be announced on Canvas

Additional Information Target Audience

Compulsory course within the Master programme Management, Policy Analysis and Entrepreneurship for the Health and Life Sciences (MPA) and the Societal differentiation of Health, Life and Natural Sciences Masters programmes

Additional Information

Attendance to training workshops / working groups is indispensable

Custom Course Registration

Please note that Workgroups are formed through Canvas at the start of the course. It is not possible to enrol for workgroups otherwise unless announced via Canvas.

Concepts in Chemical Biology

Course Code	XM_0080
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I. Drienovska
Examiner	dr. I. Drienovska
Teaching Staff	dr. S.A.K. Jongkees, dr. I. Drienovska, dr. V.U. Chukhutsina
Teaching method(s)	Written partial exam, Lecture

Course Objective

Course Objective

The ability to critically discuss published data will be trained. After the course, students are expected to know:

- the hallmarks of cancer and know the underlying principles
- core cancer signaling pathways
- inhibition strategies of signaling pathways – e.g. use of small molecules, stabilized peptides
- chemical reactions for the selective covalent modification of proteins
- bio-orthogonal and ligand directed reactions
- classical protein engineering (natural: directed evolution, disulfide…)
- Biology of the coronavirus life cycle
- SARS antiviral strategies
- (non-)natural protein modifications for improved functions
- Mechanisms of carbohydrate-active enzymes
- Chemical tools and approaches for the study of glycans

Course Content

Apart from a general introduction, four main topics are discussed, introducing different techniques central to Chemical Biology research:

- Virus Biology and Inhibition
- Chemical Glycobiology
- Targeting Hallmarks of Cancer
- Protein Engineering

Additional Information Teaching Methods

1. Lectures: Lectures will be given in English, and their content as well as the associated literature are preparation for the assignments.
2. Case Study and Assignment: The students are provided with one or more research papers. Alone or in groups of two, questions regarding these case studies are addressed in an assignment.
3. Guided Peer-to-Peer Discussions: In seminar sessions, Peer-to-Peer exchange for each topic will take place. Within these, the students are expected to exchange questions and help each other in answering these. Students are expected to participate actively. Discussions should be held in English and also a joined by one of the lecturers.

Method of Assessment

Students that have handed in all four assignments can participate in the final exam. The course grade will only be based on the grade of the exam.

Additional Information Target Audience

Recommended background knowledge: basic physico-chemical knowledge: e.g. Gibbs free energy, forces (coulomb, H-bonds, van der Waals), - basic chemical knowledge: e.g. electronegativity, bond character, acids & bases, organic synthesis - basic biochemical knowledge: e.g. general characteristics of a bacterial, mammalian and plant cell, phospholipids & biological membrane formation and function, nucleotides, DNA, RNA, double helix formation, gene regulation, amino acids and peptide bond formation, protein primary, secondary and tertiary structure, transcription, translation, post-translational modifications, signal transduction, principle of enzymatic reactions.

Additional Information

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130553

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Current Sustainable Energy Technologies

Course Code	X_422582
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.N. Frese
Examiner	dr. R.N. Frese
Teaching Staff	dr. R.N. Frese, dr. A. Baldi
Teaching method(s)	Seminar, Lecture

Course Objective

This course is about asking the right questions and obtaining reliable answers on everything related to energy usage and existing, large scale, renewable energy resources. You will read the book Renewable energy without the hot air, which provides the starting point of the energy transition we are currently bringing about. After reading and commenting on the text, you can explain the basic features of these technologies and make quantitative predictions for implementing each of these technologies. During the course, you will investigate topics of your own choice, focus on their current status, and seek ways to remove hurdles and bottlenecks. After the course, you have an overview on how to investigate the technical, economic and societal feasibilities of all possible forms of sustainable energy, including relevance and positive and negative effects.

Course Content

The course deals with all parts, Part I, II and III, of the book of MacKay, Renewable energy without the hot air, which is free to download at the book website. During the course you read the book of MacKay and annotate and

discuss parts with fellow students.

Based on this and your own interests, you focus on the current state of specific topics at general media websites and other sources, from which you formulate several questions. You research topics of interest first, and focus on specific aspects that cannot be found online and are of interest to policymakers or others involved in renewable energy. From this brief investigation, you formulate focused questions with impact and provide a hypothetical answer with only rough calculations. The hypothetical answers should contain well-argued estimates. All questions must be relevant and realistic and will as such be evaluated.

During several workgroups you will learn as a team the methodology and style of reporting, and receive feedback. Subsequently. teams present their findings and you continue to work individually on your own questions including some from a fellow student. Towards the end you will review the written work of another student, and after the feedback on your work is received, you take the comments into account and prepare the final version of your report.

Attendance to all lectures and workgroups is compulsory. All work must be submitted completely and before the deadlines in order to pass this course or to be eligible for a retake. A minimum score applies for parts of the work in order to pass the course. No parts of the course are transferrable to another year.

Additional Information Teaching Methods

This course is both team and self study based, with feedback during workgroups from students and teachers. You are expected to research topics via sources such as general news and media sites, dedicated websites for specific technologies and expert reports on implementation of technology at a specific place. You report multiple times on your research and can discuss impact of your research within workgroups and individually with teachers. An active study approach of an individual student towards asking questions and feedback is essential.

Method of Assessment

The grade is a weighted average of the following items:

1. Formulation of topical researched questions and hypothesis
2. Answers to questions using the correct procedure
3. reading the book and annotating and discussing topics
4. reviewing work of fellow students

A retake is only allowed once, and only for students that have:

1. participated in all workgroup sessions
2. read the entire book with the minimal requested annotations
3. submitted all requested questions and all answers within the deadline with relevant content
4. submitted a review containing relevant content

Note: no parts of this course is transferable to another year

Entry Requirements

This course is for energy related master programs only. Prior knowledge that is expected is general knowledge of energy and power generation, renewable energy and related issues globally. This course necessitate skills like finding reliable information on the internet, critically assessing claims related to energy and output on websites or other sources. Also students should be able to assess reliable news sites for topical aspects, quick assessing data for relevance and quick reading of reports by municipalities or organizations.

The course is open to all master students within a Chemistry, Physics or Science Business and Innovation program . As well as all students with a bachelor degree in any of these three programs. All other students must have successfully followed courses in mechanics and electricity and magnetism at university level in possession of a physics based working knowledge of energy and energy related topics.

Literature

Literature: David J.C. Mackay, 'Sustainable energy - without the hot air', 2008; available free online at ttp://www.withouthotair.com
Prior the course you can browse the book but do not read it as this is being monitored during the course and part of the grade.

Additional literature that you can read: general media websites on renewable energy, news sites, sustainability and repositories of energy data

Additional Information Target Audience

This is a full time course. Following other courses, including internship in the same period is not recommended. Note: work or parts of work cannot be transferred to another year.

Additional Information

This course is individual research driven and contains reading of the book, and reading of topics of interest on the internet or other sources. Subsequent self-formulating research questions, hypothesis and answers. Each week there are two days of three hour classes.

Explanation Canvas

On canvas most relevant information can be found in the form of examples, presentations and other docs. Students are expected to read carefully all these docs without specific instruction by a teacher. Also teachers will gradually update the canvas site, communicate to all students, specific groups or individual students. All participants are expected to actively check for messages during the course on a daily basis. Also, students can actively engage with teachers via Canvas, including continuous feedback during the course

Recommended background knowledge

This course is challenging for those that do not read much, especially no news sites, and who do not have a general working knowledge on issues related to renewable energy and sustainability in our present time. Also an active attitude towards requesting feedback in workgroups and general classes for discussion is very helpful.

Drug Target Biochemistry and Signaling

Course Code	XM_0111
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M.J. Smit
Examiner	prof. dr. M.J. Smit
Teaching Staff	prof. dr. M.J. Smit, dr. M.H. Siderius
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

Deepening of the knowledge of the biochemical aspects of drug targets and their role in cellular signaling in health and disease, with the aim to understand their importance in modern pharmacology.

Course Content

This course is designed for students with a background in the field of pharmaceutical and life sciences (or have followed Principles of Pharmaceutical Sciences) and an interest in the biological and pharmacological aspects of drug discovery. The course will focus on biochemical properties, including structures, of the main families of drug targets and their involvement in cellular signaling networks in health and disease (e.g. CNS-related disorders, cancer). Different strategies to identify, validate and modulate drug targets (e.g. ‘omics’ approaches, siRNA, CRISPR/Cas9, biologicals) will be addressed. New concepts in the architecture of signaling networks and mechanisms of signal transfer will be discussed.

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Written examination and assignments

Literature

Primary literature and handouts.
'Cell signaling', second edition. Authors: Wendell Lim and Bruce Mayer ISBN: 978-0-367-27937-0

Additional Information Target Audience

1DDS, 1BMS, MSc Bioinformatics and Systems Biology with Life Science background/BSc

Recommended background knowledge

BSc Pharmaceutical Sciences or equivalent or Principles of Pharmaceutical Sciences or BSc in Life Sciences

Entrepreneurship in Science and Technology

Course Code	XMU_0066
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. E. Masurel
Examiner	prof. dr. E. Masurel
Teaching Staff	prof. dr. D. Iannuzzi, prof. dr. E. Masurel
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130496

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

High-Throughput Screening

Course Code	X_435047
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J. Kool
Examiner	dr. J. Kool
Teaching Staff	dr. J. Kool
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130524

Additional Information Teaching Methods

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Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Management of Sustainable Innovation

Course Code	E_DBI_MANSI
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	D. Diriker
Examiner	D. Diriker
Teaching Staff	D. Diriker
Teaching method(s)	Study Group, Lecture

Course Objective

Bridging Theory & Practice (knowledge):
Understand and critically assess theories of organizing for sustainable innovation and understand how entrepreneurs , business leaders, governments, and citizens can help to co-create a more sustainable future and how digitalization might be leveraged to support this future.

Bridging Theory and Practice (application):
Apply theoretical insights to conceive new approaches to organizing the implementation of sustainable innovations.

Professional & Social skills:
Ability to work effectively in a team. Have a good contribution, content and process wise: Pro-active teamwork, timely, friendly and constructive.

Course Content

Sustainability is now at the core of our societies and economies: governments, companies and citizens join forces to fight global warming, plastic pollution, and other negative developments. Rapid changes due to digitalization have further unleashed new forms of organizing that enable and accelerate coordination, collaboration, and knowledge efficiencies.

Despite the promise of digitalization and sustainability imperatives for transforming business and society, a strategic focus on convergence of these imperatives as a means for organizing and accelerating the development and implementation of sustainable innovations is relatively new for both practice and scholarship.

In this course, we explore how business leaders and entrepreneurs around the world do things differently, to radically change the way they do business, what role they want to play in our societies, and leverage digital technologies to overcome barriers to sustainable development. You will apply theories on sustainable innovation, digital innovation, innovation processes to examine contemporary cases and learn how to organize for sustainable innovation by mobilizing and integrating information, knowledge and critical resources of diverse stakeholders.

Additional Information Teaching Methods

Lectures: Interactive Lectures (Online & In-person); Guest Lectures Tutorials: Interactive discussion and case work (Online & In-person)

Method of Assessment

Individual written exam.
Group assignment (exact assignment will be announced later).
(Peer) evaluation of contribution to the course and group

Entry Requirements

For the course it is required that students know:

- The concept of Triple P: Planet, people, profit.
- The Brundtland report Our common future 1987
- The 17 UN Global Sustainability goals.

Literature

Academic articles (to be announced in course manual).

All lectures will discuss 2-3 academic articles that will help students to understand sustainable innovations and the organizing processes that are necessary to create and implement these.

Book for background reading:
Senge, Peter M., Bryan Smith, Nina Kruschwitz, Joe Laur, Sara Schley, 2008, The Necessary Revolution: How Individuals and Organizations Are Working Together to Create a Sustainable World, Doubleday (or later edition)

Additional Information Target Audience

Students that want to make a change in existing companies, or co-create change in our societies. This is a course in which students are requested to participate very actively! Students that do not have sufficient time or motivation, are advised not to follow this course.

Concrete profile:

- Students that want to become business or policy consultant in sustainability.
- Students that want to work in a community or province to develop knowledge on how regions can be transformed to co-create value in business and society.
- Students that want become social entreprenuers or implement sustainable innovations in their future careers

Recommended background knowledge

Knowledge of innovation management, diffusion of innovation.
Please note: This course is not about innovation (technical) but about the management of innovation, e.g. stakeholder management, co-creation, open innovation, sustainable entrepreneurship.

Molecular Modelling

Course Code	XMU_0059
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. C. Fonseca Guerra
Examiner	prof. dr. C. Fonseca Guerra
Teaching Staff	dr. T.A. Hamlin, prof. dr. C. Fonseca Guerra, prof. dr. F.M. Bickelhaupt
Teaching method(s)	Seminar, Other, Practical, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130572

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Photosynthesis and Energy

Course Code	X_422553
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. V.U. Chukhutsina
Examiner	dr. I.H.M. van Stokkum
Teaching Staff	dr. V.U. Chukhutsina, dr. I.H.M. van Stokkum
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130439

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Physical Organic Chemistry

Course Code	X_435663
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. T.A. Hamlin
Examiner	prof. dr. F.M. Bickelhaupt
Teaching Staff	dr. T.A. Hamlin, prof. dr. C. Fonseca Guerra, dr. T. Hansen MSc, prof. dr. F.M. Bickelhaupt
Teaching method(s)	Seminar, Lecture

Course Objective

The student will be introduced to the fundamental principles of physical organic chemistry that dictate the reactivity of molecules.

The student will develop a deep understanding of how rates of reactions can be altered and the role of catalysts in accelerating reactions.

The student will be exposed to foundational concepts such as acid/base catalysis, carbocation/carbanion chemistry, chemical bonding, stereochemistry, isotope effects, reaction mechanisms, aromaticity, and various types of organic reactions like nucleophilic substitutions/eliminations and pericyclic reactions.

The student will be equipped with skills to explore a wide range of fields, spanning the range from fundamental organic reactions to supramolecular (biomolecular) interactions and recognition.

Course Content

This Physical Organic Chemistry course aims to cultivate critical thinking and problem-solving skills. Students will delve into the fundamental principles governing molecular reactivity, exploring topics such as catalyst-driven reaction rate modifications, acid/base catalysis, carbocation/carbanion chemistry, chemical bonding, stereochemistry, isotope effects, and reaction mechanisms. Through hands-on exercises, students will learn to formulate arrow-pushing mechanisms for experimental observations, assess existing theories, design validating experiments, and evaluate alternative explanations. The course equips students with the tools to navigate a wide range of fields, from foundational organic reactions to the complexities of supramolecular interactions and biomolecular recognition, fostering a deep understanding of the dynamic world of chemical reactions and molecular interactions.

Additional Information Teaching Methods

The course is comprised of lectures, self-study, and tutorial sessions.

Method of Assessment

Written examination.

Entry Requirements

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130522

Literature

BSc Chemistry

Additional Information Target Audience

Students interested in the topic of physical organic chemistry.

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Plastic and Chemical Pollution

Course Code	AM_1259
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.H. Brandsma
Examiner	dr. S.H. Brandsma
Teaching Staff	dr. S.H. Brandsma, dr. R. Oldenkamp, dr. L.A. Baumann, dr. S.L. Hughes, dr. I. van der Veen
Teaching method(s)	Computer lab, Seminar, Study Group, Symposium, College case, Practical, Lecture

Course Objective

Objectives

- To evaluate key environmental pollutants, their sources, properties, distribution and fate in the environment.
- To demonstrate quantitative information on the effects of pollutants on ecological and human health.
- To demonstrate how environmental risks of pollutants are assessed.
- To relate the concept of systems thinking, and its application to studying and understanding environmental pollution problems.
- To describe the key principles of environmental science that underpin both the assessment and governance of plastic and chemical pollution.
- Formulate the environmental risk of a chemical pollutant in a report and an oral presentation.

Contents

This course provides an introduction to environmental chemistry and (eco)toxicology of plastic and chemical environmental pollution, one of the hallmarks of the hotly-debated ‘Anthropocene’, the human-dominated epoch. Is this environmental pollution just a global bundle of bad news, or can/does science and society adequately respond? If you are interested in learning about plastic pollution and other hot chemical pollution topics this course is for you.

Course Content

Topics include:

- Sources and emissions and spatial transport of plastic and chemical pollution
- Properties affecting transport and bioaccumulation of plastic and chemical pollutants
- Bioaccumulation and biotransformation in living organisms
- Ecotoxicology and human toxicology of selected plastic and chemical pollutants
- Risk assessment of plastic and chemical pollutants
- Systems thinking in environmental chemistry and one world chemistry
- Societal and policy responses to address pollution

Plastic and chemical pollution is dispersed throughout our environment, including food chains and human bodies, making it a problem of global proportions. But a solvable problem. It may be surprising but most of the >100,000 chemicals synthesized in the past 70 years have not undergone rigorous safety testing before entering the marketplace, and many of these are high-production volume chemicals, including polymers.

Approximately 80% of today’s chemicals sector produces plastic and polymers. Past lessons with pollutants like dichlorodiphenyltrichloroethane (DDT), polychlorinated biphenyls (PCBs), and chlorofluorocarbons (CFCs) showed us that chemicals with economic and societal benefits can have long-lasting serious negative consequences for public health, ecosystems, as well as the economy. After decades of research on an often chemical-by-chemical basis, science has been revealing how certain chemicals act as developmental neurotoxicants, endocrine disruptors, genotoxicants or immunotoxicants. And this knowledge is powerful. Such research is instrumental in evidence-based, risk-based policymaking that phased out many persistent organic pollutants to date. Cutting-edge analytical environmental chemistry is critical for first signaling and then quantifying ‘emerging’ pollutants in the environment, such as microplastics, endocrine disrupting chemicals (incl. plastic additive chemicals), perfluorinated chemicals, chlorinated paraffins, etc. Without analytical data, there’s no story to tell. Understanding the toxicity of the chemicals to humans and ecosystems gives meaning to the exposure concentrations measured in the field and laboratory. These data are essential input to environmental and human health risk assessments, policy-making and legislation, and design choices during manufacturing (e.g. green chemistry substitutions and functional alternatives). The societal and policy responses pave the way for elimination of pollution sources and the stimulation of innovations that act to prevent pollution, improving the quality of the living environment for all. This course may stimulate you to think about whether or not we currently are using our best science in creating the chemical-material world in which we are living.

Additional Information Teaching Methods

Teaching method

- Lectures
- Seminars
- Presentation
- Self-study
- Working on a written project
- Supervision/feedback

Method of Assessment

The final grade is calculated from the grade of the exam (65%), assignment report (25%), and oral presentation

(10%). Minimum grades needed are: exam 5.0, report and presentation 5.5. In case resubmission of the report is needed a maximum grade of 6.0 will be assigned.

Entry Requirements

BSc degree in either hydrology, ecology, environmental sciences, chemistry, biomedical sciences, or public health.

If you have already followed the BSc Analytical and Environmental Chemistry (AEC) course (51128ANM6Y) then be aware that there can be overlap between some of the lectures given in this course.

Literature

Literature

“Environmental toxicology, an open online textbook”

Other: Additional material available on the Canvas site.

Additional Information Target Audience

Is part of/may be part of/is open to students of

- MSc Ecology (elective)
- MSc Hydrology (elective)
- MSc Earth Sciences – track Global Environmental Change and Policy (elective)
- Earth Sciences’ Earth & Climate stream (elective)
- MSc Biomedical Sciences (elective)
- MSc Chemistry (elective)

Additional Information

Attendance at lectures is highly recommended and is mandatory for the presentations.

The course is given at the VU. The first 4 weeks of the course will cover most of the new material, the second half of the course will be more focused on assignment, readings, a symposium for presentations of the assignment work, and finally the exam. Ecology students may complete work on assignment in the first four weeks, with only their presentation and exam in the second half of the course schedule.

Recommended background knowledge

BSc degree in either hydrology, ecology, environmental sciences, chemistry, biomedical sciences, or public health.

If you have already followed the BSc Analytical and Environmental Chemistry (AEC) course (51128ANM6Y) then be aware that there can be overlap between some of the lectures given in this course.

Protein Analysis

Course Code	X_435045
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Bärenfänger
Examiner	dr. J.M. Bärenfänger
Teaching Staff	dr. K. Jooß MSc, dr. J.M. Bärenfänger
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130525

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Relativistic Quantum Chemistry

Course Code	XMU_0029
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. L. Visscher
Examiner	prof. dr. L. Visscher
Teaching Staff	prof. dr. L. Visscher
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130547

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Science and Communication

Course Code	AM_470587
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. C. Boersma MA MSc.
Examiner	dr. C. Boersma MA MSc.
Teaching Staff	dr. C. Boersma MA MSc., T.F.L. Roedema MSc, L.B. Clauss MSc, K.E. De Long MSc
Teaching method(s)	Study Group, Lecture

Course Objective

- Gain theoretical insight in the nature of science,
- Gain theoretical insight in the nature of communication,
- Gain theoretical insight in the relationship between science and society,
- Gain insight in the role of science communication in this relationship,
- Acquire knowledge of different theories and models of science communication,
- Acquire knowledge of different strategies, media and activities for science communication,
- Learn how to practically apply theoretical concepts from the field of science communication in communicating science,
- Develop practical skills for science communication (especially writing and giving oral presentations).
- Reflect on your own knowledge and competencies pertinent to your projected (ideal) role as science communicator.

Course Content

Science is all around us and shapes our lives in many different ways. From the vaccines you need to get when traveling abroad to the smartphone you use on a daily basis, and from the public transportation you use to get to the university to the ingredients of your toothpaste: scientific knowledge is elemental to all of these. Simultaneously, society shapes the ways in which science and technology develop too. Science, technology and society influence each other continuously—or, to put it differently, they ‘communicate’.

Students of the Science Communication specialization are expected to become experts in understanding and designing interactions between science and society. In order to make this interaction fruitful and valuable for both science and society, it is first of all important to gain theoretical knowledge about science, about communication and about science communication. Science and Communication provides students with the theoretical and conceptual foundations of the discipline of science communication. Thus, you will develop an in-depth understanding of communication processes at the core of several interfaces, including those between scientists from different disciplines, between different sciences and their stakeholders, and between science and the public.

Additional Information Teaching Methods

Expected contact hours:
Lectures (20 h)
Workgroups (25 h)
Home-study for group assignments (15 h)
Home-study for individual assignments/exam (100 h)

Method of Assessment

a) Participation. (pass/fail)
This consists of the following:
- the 'CARQ' presentations and summaries;
- Active participation in the work groups.
All these are assessed as pass or fail. For each one you fail, you have to do an alternative assignment.
Nota bene: if you fail your participation, this cannot be compensated with an alternative assignment!

b) An interview with a science communicator of your own choosing, including a reflection report. (20%)
d) "TED-talk" in which you present the research you did (e.g. for your Bsc thesis or (first) Msc internship) and reflect on it. (20%)
e) Exam. (60%)

To pass, your grades for the assignments and exam all need to be 5.5 or higher.

Resit: In case your grade is not sufficient, you have to take a resit. This can either consist of a second attempt at (b), (c) or (d), or a re-exam.

Literature

Academic articles. Direct links to articles will be provided on Canvas.

Additional Information Target Audience

The course Science and Communication is a compulsory course for students of the Master specialisation Science Communication (Wetenschapscommunicatie) and is a prerequisite for the internship. Science and Communication is an optional course for students from other master programs in the health and life sciences.

Science and Society in Historical Perspective

Course Code	X_400424
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G.J. Somsen
Examiner	dr. G.J. Somsen
Teaching Staff	dr. G.J. Somsen, dr. D.J. Beckers
Teaching method(s)	Seminar, Lecture

Course Objective

To understand how science and society have co-evolved and to know how to research this co-evolution.

Course Content

We are used to thinking of science as the prime shaper of our modern world, but how exactly does that work? How have the sciences come to play this role in our society and, reversely, how have politics and culture impacted the scientific enterprise? In this course we will study the changing relationship between science and society over the centuries by looking at a series of themes. We will examine how scientists have dealt with the public, we will study science's relationship to religion, and we will ask how Western science really is. To address these questions, we will use historical literature, videos and podcasts as well as original source texts and pictures. We will also try our own hand at historical research, through a number of skills trainings and a final paper.

Additional Information Teaching Methods

Seminars (2hrs per week), skills trainings (twice), lectures (occasionally), presentations ands feedback.

Method of Assessment

Assessment is based on active participation during the seminar (10%) and a piece of historical research, reported on in a presentation (20%) and a final paper (70%).

Entry Requirements

Bachelor degree

Literature

Available via Canvas

Additional Information Target Audience

- Master students in the sciences interested in historical reflection;
- Master students in the humanities interested in understanding the role of science in society.

Additional Information

More information with the course coordinator: Geert Somsen, Athena Institute, g.j.somsen@vu.nl

Recommended background knowledge

None necessary, except what you have learned in your BA.

Science in Dialogue

Course Code	AM_1002
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	dr. E. Gommeh
Examiner	dr. E. Gommeh
Teaching Staff	mr. D.T. van der Lugt MSc, P.E. van der Aar, M.C. Vos MA MSc., J. Putter MSc, A.M. Aris, A.R. de Leeuw
Teaching method(s)	Seminar, Study Group, Written partial exam, Lecture

Course Objective

To gain knowledge of and insight into:

- the basic concepts and issues in understanding science-society interactions and interfaces, both from a science and technology studies and communication science perspective
- the nature and course of interpersonal and group communication processes relevant to the formal and informal dialogue between science and society
- the nature and form of dialogical science communication, aimed at reflective learning and mutual understanding

To acquire or improve:

- individual skills for effective interpersonal communication
- individual skills for the design and facilitation of science-society dialogue

Course Content

This course invites you to explore the democracy of science and technology with a particular focus on the communicative aspects of a fruitful science - society dialogue. Science, technology and innovation hold a great promise for the progress of our societies. Take for example the opportunities of nanomedicine for targeted drug delivery or synthetic biology for renewably energy or the recycling of plastic waste. At the same time these developments are potentially controversial. They lead to a variety of questions and concerns related to risks, benefits and wider moral issues. For example, nanotechnology may contain risks for health and environment. Synthetic biology may radically change the nature and meaning of life.

Clearly, advances in science do not always match the needs, desires and expectations of stakeholders and citizens in society. On the other hand, parts of society might not always appreciate the nature of scientific findings and what they have to offer. For a fruitful relationship between science and society – especially regarding topics that are potentially controversial – a constructive science-society dialogue is necessary.

This course offers advanced lectures on the basic concepts and theories about the dynamic relationship of science and society, the nature of societal controversies surrounding science, technology and innovation, and the basic concepts and issues of dialogical science communication: communication, learning, dialogue, understanding, controversy, and democracy. A series of seminars and group assignments present communicative tools and interaction spaces that can be used to design and facilitate science-society interactions. Through training workshops you will focus on improving your individual communication and facilitation skills.

Additional Information Teaching Methods

Lectures (14h), Workgroups (28h), Training workshops (24h), Dialogue presentations (12h), Selfstudy (remaining hours)

Method of Assessment

Group assignment (50%), Two digital exams (30%), Reflection report (20%).

All assignments must be passed (grade > 6).

Literature

Is announced on Canvas one month before start of the course

Additional Information Target Audience

Optional course in the MSc specialization Science Communication

Additional Information

Independence and a cooperative attitude is expected. Attendance to workgroup/ seminars and training sessions is

mandatory.

Scientific Computing and Programming

Course Code	X_435076
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. L. Visscher
Examiner	prof. dr. L. Visscher
Teaching Staff	prof. dr. L. Visscher
Teaching method(s)	Practical, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130526

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Scientific Writing in English

Course Code	X_400592
Credits	3
Period	P3, P6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	B.I.M. Schouten
Examiner	B.I.M. Schouten
Teaching Staff	
Teaching method(s)	Study Group

Course Objective

The aim of this course is to provide Master's students with the essential linguistic know-how for writing a scientific article in English that is well organized, idiomatically and stylistically appropriate and grammatically correct.
At the end of the course students:

- know how to structure a scientific article;
- know what the information elements are in parts of their scientific article;
- know how to produce clear and well-structured texts on complex subjects;
- know how to cite sources effectively;
- know how to write well-structured and coherent paragraphs;
- know how to construct effective sentences;
- know what collocations are and how to use them appropriately;
- know how to adopt the right style (formal style, cohesive style, conciseness, hedging)
- know how to avoid the pitfalls of English grammar;

- know how to use punctuation marks correctly;
- know what their own strengths and weaknesses are in writing;
- know how to give effective peer feedback.

Final texts may contain occasional spelling, grammatical or word choice errors, but these will not distract from the general effectiveness of the text.

Course Content

The course will start with a general introduction to scientific writing in English. Taking a top-down approach, we will then analyse the structure of a scientific article in more detail. As we examine each section of an article, we will peel back the layers and discover how paragraphs are structured, what tools are available to ensure coherence within and among paragraphs, how to write effective and grammatically correct sentences and how to choose words carefully and use them effectively.

Topics addressed during the course include the following:

- Structuring a scientific article
- Considering reading strategies: who is your readership? How do they read your text? What do they expect? How does that affect your writing?
- Writing well-structured and coherent paragraphs
- Composing effective sentences (sophisticated word order, information distribution).
- Academic tone and style: hedging – why, how, where?
- Using the passive effectively
- Understanding grammar (tenses, word order, etc.)
- Understanding punctuation
- Referring to sources: summarising, paraphrasing, quoting (how and when?)
- Avoiding plagiarism
- Vocabulary development: using appropriate vocabulary and collocations

Additional Information Teaching Methods

Scientific Writing in English is an eight-week course and consists of 2 contact hours a week. Students are required to spend at least 6 to 8 hours of homework per week. They will work through a phased series of exercises that conclude with the requirement to write several text parts (Introduction, Methods, Discussion and Abstract). Feedback on the writing assignments is given by the course teacher and by peers.

Method of Assessment

Students will receive the three course credits when they meet the following requirements:

- Students hand in three writing assignments (Introduction, Methods, Discussion)
- Students get a pass mark for all writing assignments;
- Students provide elaborate peer feedback (Introduction, Methods, Discussion, Abstract);
- Students attend at least 7 out of 8 sessions;
- Students are well prepared for each session (i.e. study the theory for each session and participate in online discussions about the theory);
- Students participate actively in class and respect deadlines;
- Students do not plagiarise or self-plagiarise.

Writing assignments:

- If students have a BSc thesis in a traditional thesis form (e.g., 20+ pages) and written in English, they may use this for the writing assignments.
- If students have a BSc thesis in a traditional form (e.g., 20+ pages) written in another language than English, they may use this for the writing assignments.
- If students have written a paper or report in English that’s not already in article form, they may use this for the writing assignment.
- If students are working on their MSc thesis or internship report when taking Scientific Writing in English, they may use this for the writing assignments. They will have to notify their supervisor to make sure that they won’t be accused of self-plagiarism.
- If students cannot or do not wish to use any of the above-mentioned texts for the writing assignments (1-4), they are expected to do a limited Literature Review on a topic in their field of research, using at least 5 articles.

Students are not allowed to use the following texts for the writing assignments:

- A BSc thesis written in English that’s already in article form.
- A MSc thesis written in English that’s already in article form (and that has already been marked).

- An internship report written in English that’s already in article form (and that has already been marked).
- A paper or report written in English that’s already in article form.

Literature

Effective Scientific Writing: An Advanced Learner's guide to Better English, 5th edition (May 2017) (A. Bolt & W. Bruins, ISBN 978 90 8659 617 1). VU bookstore: €34.50

Additional Information Target Audience

This course is only open to students of the two-year Master's programmes of the Faculty of Science. These students are only eligible to the course if they have already conducted scientific research (e.g. for their Bachelor’s thesis) or if they will be working on a research project when taking Scientific Writing in English.

Additional Information

To do well, students are expected to attend all lessons. Group schedules are to be found at rooster.vu.nl and on Canvas.

A VUnet registration for this course automatically gives access to the corresponding Canvas site. Group registration only takes place via Canvas.

Make sure Scientific Writing in English does not overlap with another course.

If you have registered for a group in Canvas, you are expected to attend all sessions (eight). If you decide to withdraw from the course, do so in time in VUnet. This will avoid a 'fail' on your grade list for not taking part in this course and allows other students to fill in a possible very wanted group spot.

Full time students with their main registration at VU will be given preferential treatment for placement in this course. For secondary students proof of enrollment is not a guarantee of placement.

Custom Course Registration

Important: each group has a minimum of 18 and maximum of 24 participants, so students should register on time through VUnet to ensure a place in one of the (designated) groups. If you have registered for a group in VUnet, you are expected to attend all sessions (eight). If you decide to withdraw from the course, please do so in time. This allows other students to fill in a possible very wanted group spot.

Second Quantization Techniques in Quantum Chemistry

Course Code	XMU_0030
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. L. Visscher
Examiner	prof. dr. L. Visscher
Teaching Staff	prof. dr. L. Visscher
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130548

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Separation Sciences

Course Code	X_435609
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. G.W. Somsen
Examiner	prof. dr. G.W. Somsen
Teaching Staff	prof. dr. G.W. Somsen
Teaching method(s)	Seminar, Lecture

Course Objective

This is a course in the VU-UvA joint degree Chemistry. See UvA study guide for goal and learning objectives (link below).

Course Content

This is a course in the VU-UvA joint degree Chemistry. See UvA study guide for course content:
https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130517

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Spectroscopic Analysis

Course Code	XMU_0039
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. A.M. Rijs
Examiner	
Teaching Staff	dr. F. Ariese, prof. dr. A.M. Rijs
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130558

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Synthetic Strategies

Course Code	XMU_0016
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. E. Ruijter
Examiner	
Teaching Staff	dr. T. Hansen MSc, prof. dr. E. Ruijter
Teaching method(s)	Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130549

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Teaching Assistant

Course Code	XM_432741
Credits	3
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. J.E. van Muijlwijk-Koezen
Examiner	prof. dr. J.E. van Muijlwijk-Koezen
Teaching Staff	
Teaching method(s)	

Course Objective

The main goal is to improve your skills in supervision and teaching through work in a (practical) course, including familiarising yourself with the specific STEM didactics.

Course Content

Teaching skills are practised and developed during a (practical) course and assessed through both a performance evaluation and a written reflection. Students are recommended to follow the course 'Tutoring Students' prior for a stronger theoretical basis. There is also additional required material on Canvas for students who work in wet labs.

Additional Information Teaching Methods

Hands on course:

You will become a member of the team of supervisors for practical courses or working classes for undergraduates and assist in the lab or classroom. With the aid of feedback, you will improve via learning by creating goals, taking (teaching) actions to develop these goals, and reflecting on the success of these actions in a final report.

Method of Assessment

Execution during the course (50%) and concluding reflective assignment (50%). Both must be successfully completed.

Additional Information Target Audience

mDDS, mChem, mPhys, mMath, mMNS, mSBI

Additional Information

Limited spots available. This course can be done in any period during the year. Make sure you arrange everything well in advance (see also Canvas page).

Find a coordinator/teacher of a (practical) course that needs a student assistant for guiding a practical course or tutorials. You are responsible for finding a Student-as-Teacher position yourself. The duration and the number of hours required to supervise in this assistantship will determine whether you are eligible for the 3 EC or 6 EC variant of this course.

Custom Course Registration

Contact your master coordinator IN TIME. The earlier the better to ensure that this fits in your Personal Education Plan.

Explanation Canvas

All information is on Canvas and it is essential that students take responsibility for taking initiative, following the provided instructions, and contacting the relevant teachers on time.

Recommended background knowledge

X_432625, period 3, Tutoring Students or equivalent via Center for Teaching & Learning

Teaching Assistant

Course Code	XM_432742
Credits	6
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. J.E. van Muijlwijk-Koezen
Examiner	prof. dr. J.E. van Muijlwijk-Koezen
Teaching Staff	
Teaching method(s)	

Course Objective

The main goal is to improve your teaching skills and get familiar with the specific pharmaceutical sciences/chemistry didactics (Science students as teachers)

Course Content

Practice during a (practical) course supervised by the course lecturer and a written reflection. For assistance during practical work the course 'tutoring students' is a requirement.

Additional Information Teaching Methods

hands on course:
You will become a member of the team of supervisors for practical courses or working classes for undergraduates and assist in the lab or classroom. With the aid of feedback and intervision, you will improve via learning by doing. The theoretical background has been taught in the course 'tutoring students'

Method of Assessment

Execution during the course and concluding reflective session

Additional Information Target Audience

mDDS, mChem, mPhys, mMath, mMNS, mSBI

Additional Information

Limited seats available, fixed periods

Find a coordinator/teacher of a (practical) course that needs a student assistant for guiding a practical course or tutorials. You are responsible for finding a student assistant position yourself. The duration and the number of hours required to supervise in this assistantship will determine whether you are eligible for the 3 EC or 6 EC variant of the Teaching assistant course. Make sure this is agreed upon beforehand! The teaching assistant course can be done in any period during the year. Make sure you arrange everything well in advance (see also Canvas page).

Custom Course Registration

Contact your master coordinator IN TIME. The earlier the better.

Recommended background knowledge

X_432625, period 2, tutoring students or equivalent via Center for Teaching & Learning

Tutoring Students

Course Code	X_432625
Credits	3
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	I.T. Braadbaart MA
Examiner	prof. dr. J.E. van Muijlwijk-Koezen
Teaching Staff	dr. M. Wijtmans, prof. dr. J.E. van Muijlwijk-Koezen, I.T. Braadbaart MA, dr. D.J. Scholten
Teaching method(s)	Lecture

Course Objective

This course aims to prepare students for coaching tasks in tutorials and practical courses. Students will explore different aspects of teacher-student interaction, and by the end of the course will be able to reflect critically on their beliefs and practices about teaching, guide pre- and post-experiment discussions, support students in problem-solving, plan for safety in practical work, and give written feedback.

Course Content

The course asks students to engage in literature, discussion and practical simulations to support students' understanding of appropriate teaching and coaching practices in tutorials and practical courses. Students will analyse various learning aims within a laboratory setting and prepare, conduct and reflect on a presentation of a pre- and post-experiment discussion in tutorials and practical courses. They will observe and interpret the application of problem solving and coaching models in tutorials and practical courses. Attention will be paid to strengths and weaknesses in models of teacher-student interaction, with a focus on questioning. An important constituent is the student's analysis of and reflection on his/her own teaching approach. Topics on safety and lab journal procedures in practical courses as well as on the grading of lab reports are also included.

Additional Information Teaching Methods

3,5 consecutive hours per session (2x per week over four weeks):

- Lectures and whole class discussions
- Hands-on practice moments
- Self-study
- Group work

Method of Assessment

A collection of written group assignments completed throughout the course, including:

- a summary of good guiding practices
- two lesson plan designs
- a plan for pre- and post-experiment discussion
- a rubric

A collection of individual assignments completed throughout the course, including:

- a reflection on previous learning experiences
- a reflection on pre- and post-experiment discussion
- a reflection on attitudes towards guided- and open-inquiry labs
- an analysis of various strategies to repair student misconceptions in the sciences
- a reflection and analysis of problem-solving approaches
- a final critical reflection on your development throughout the course, accompanied by a formative presentation and feedback session

Literature

Will be provided.

Additional Information Target Audience

mDDS, mChem, mPhys, mMath, mMNS, mSBI

Additional Information

This course is recommendable for any MSc student who has a general interest in educational coaching strategies and models. Specific attention is given to guiding experimental / laboratory work and problem-solving skills.

Number of participants is limited to 24 (first-come, first-serve basis).

Custom Course Registration

VUnet

Explanation Canvas

Canvas will be used to share literature, submit assignments, and for other relevant course communications.

Understanding Quantum Chemistry

Course Code	X_422557

Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. L. Visscher
Examiner	prof. dr. L. Visscher
Teaching Staff	dr. K.J.H. Giesbertz-von Stein, prof. dr. L. Visscher
Teaching method(s)	Seminar, Lecture

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130531

Additional Information Teaching Methods

Method of Assessment

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.